# Functional Dependencies

The basis for BCNF & 3NF definitions

# Roadmap

- Define FDs.
- Define closures to find all FDs.
- Define superkeys to determine what should be key.
- Apply definitions to BCNF & 3NF.
- Glimpse at what's beyond BCNF.

# FD intuition

- Functional dependency $A \rightarrow B$

If you know value of $A$, that tells you value of $B$.

| A | ... | B | ... |
|---|---|---|---|
| Fred | | 2 | |
| Mary | | 7 | |
| Sam | | 2 | |
| Pat | | 1 | |

# FD – definition

Let $A$, $B$ be sets of attributes. Let $R$ be a relation.

$A$ functionally determines $B$ ($A \rightarrow B$) holds on $R$
iff
for all valid tuples $t_1$, $t_2$ in $R$,
if $t_1[A] = t_2[A]$, then $t_1[B] = t_2[B]$

$A \rightarrow B$ is a functional dependency.

# FD – an illustration

# Which FDs hold?

| emp_id | name | phone | position |
|---|---|---|---|
| E0045 | Smith | 1234 | Clerk |
| E3542 | Mike | 9876 | Salesrep |
| E1111 | Smith | 9876 | Salesrep |
| E9999 | Mary | 1234 | Lawyer |

{position} → {phone}

{phone} → {position}

# Activity – Find FDs in this instance

| a | b | c | d | e |
|---|---|---|---|---|
| 1 | 2 | 4 | 3 | 6 |
| 3 | 2 | 5 | 1 | 8 |
| 1 | 4 | 4 | 5 | 7 |
| 1 | 2 | 4 | 3 | 6 |
| 3 | 2 | 5 | 1 | 8 |

Find at least *three* FDs which hold on this instance:

- { } → { }
- { } → { }
- { } → { }

# Details

- $\emptyset \rightarrow A$
  - "Given no info, we know values of $A$."
  - I.e., constants. Typically not stored in DB.
- $A \rightarrow \emptyset$
  - "Given values of $A$, then among the things we know are the values of no attributes."
  - Always true, but useless. Example of a *trivial* FD.

# How FDs are obtained & used

- Primarily interested in FDs that hold on all instances
  - But can't simply enumerate all instances.
  - Such FDs must come from requirements analysis.

- Given $F$, what other FDs hold?

- Given $F$ & $R$,
  - Which FDs in $F$ hold? Does $R$ satisfy $F$?
  - What are $R$'s keys?
  - Is $R$ in a normal form? Put $R$ in a normal form.

$F$ = set of FDs
$R$ = relation